

A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-106660295})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K , the normalized generator σ_x , the pair (a', J) representing the torsion class e_j , the auxiliary ideal I' , the compactly represented element t , a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$I'^{(1-\sigma_x)^2}(t) J \mathcal{O}_{L_x} = \mathcal{O}_{L_x}, \quad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$ -basis of the ideal. For example, the ideal $J = \begin{pmatrix} 3459 & 2637 \\ 0 & 1 \end{pmatrix}$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z} 3459 + \mathbf{Z} (2637 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Which integral basis. The one PARI returns for the field in question – and that basis is LLL-reduced, so it is not canonical: another machine may reduce differently, and the same coordinate vector would then denote a different algebraic number. The certificate therefore records the basis it was written against, as its last field per entry and in the header for K . The verifier expresses the stored basis in its own, checks that the resulting matrix is integral with determinant ± 1 – so that both bases span the same ring of integers – and converts the coordinates before checking anything. Where the two bases agree, that matrix is the identity and nothing changes.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\text{Cl}(K)$ (as HNF ideals). A character x on $\text{Cl}(K)/5$ is recorded by its value vector $(x(\bar{\kappa}_1), x(\bar{\kappa}_2), x(\bar{\kappa}_3))$; thus $(1, 0, 0)$ is the character dual to $\bar{\kappa}_1$. The column number j records which basis element e_j of $\text{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x . Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x .

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 - s + 26665074 = 0$, of discriminant -106660295 . Class group invariants $(110 \ 10 \ 10)$, class number 11000; the three stored generator ideals (HNF) are $\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 192 & 149 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 104 & 101 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a , column e_1

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 1881y^3 + (36s - 18)y^2 - 2214820y + (-3550s + 1775)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 3762y^8 - 891479y^6 + 26225782740y^4 - 1910165218100y^2 + 336046591934375$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ -2 \ 0 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{628074433}{495168272762461299} + \left(\frac{61462}{495168272762461299}\right)s, \quad J = \begin{pmatrix} 3459 & 2637 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3459$. **Checked:** $(a')J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $21808248011093 = 757 \cdot 809 \cdot 35610361$:

$$\begin{pmatrix} 21808248011093 & 5834823254305 & 21133629511890 & 10464580457876 & 4163658108515 & 6680979443504 & 400281690000 & 19658449438256 & 18320021830249 & 1847010098211 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 2 factors with signed exponents of absolute value up to 2; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 1 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 2: character a , column e_2

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 1881y^3 + (36s - 18)y^2 - 2214820y + (-3550s + 1775)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 3762y^8 - 891479y^6 + 26225782740y^4 - 1910165218100y^2 + 336046591934375$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ -2 \ 0 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{7063831}{103621065612199888} + \left(\frac{88117}{207242131224399776}\right)s, \quad J = \begin{pmatrix} 2906 & 1111 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2906$. **Checked:** $(a')^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $48535752789461 = 7 \cdot 17 \cdot 359 \cdot 1136109941$:

$$\begin{pmatrix} 48535752789461 & 20684831812230 & 41919225679231 & 22233634708063 & 432666772455 & 34560384028454 & 18999763034252 & 234980359461 & 43608127528486 & 8578710155817 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 7 factors with signed exponents of absolute value up to 1; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \ 1 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 3: character a , column e_3

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 1881y^3 + (36s - 18)y^2 - 2214820y + (-3550s + 1775)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 3762y^8 - 891479y^6 + 26225782740y^4 - 1910165218100y^2 + 336046591934375$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ -2 \ 0 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{209517721}{100421421980737824} + \left(-\frac{46037}{100421421980737824}\right)s, \quad J = \begin{pmatrix} 2514 & 1788 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2514$. **Checked:** $(a')J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $43529656246177 = 17 \cdot 89 \cdot 28770427129$:

$$\begin{pmatrix} 43529656246177 & 23760474300758 & 26977823591851 & 25977725701912 & 7835074577365 & 21096205842733 & 29843910486809 & 7739060751154 & 21452112775666 & 8217485668397 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 2 factors with signed exponents of absolute value up to 1; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \ 1 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 4: character b , column e_1

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (s - 877)y^3 + (-12s - 96416)y^2 + (-415s + 375030)y + (-1650s + 11227340)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned}
& y^{10} - 4y^9 - 1749y^8 - 189338y^7 + 28568659y^6 - 449980264y^5 \\
& - 9697008411y^4 + 85627040190y^3 + 3623855094940y^2 + 44933719337 \\
& 800y + 198630302329600
\end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 0 \ -1 \ 0 \ -1 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{628074433}{495168272762461299} + \left(\frac{61462}{495168272762461299}\right)s, \quad J = \begin{pmatrix} 3459 & 2637 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3459$. **Checked:** $(a') J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $3864710714368 = 2^{10} \cdot 7^3 \cdot 19 \cdot 761^2$:

$$\begin{pmatrix}
809704 & 681856 & 3052 & 601944 & 597057 & 323858 & 391118 & 694745 & 84968 & 167325 \\
0 & 21308 & 19264 & 21280 & 11396 & 6240 & 21075 & 20850 & 57 & 13697 \\
0 & 0 & 4 & 0 & 0 & 0 & 2 & 1 & 3 & 2 \\
0 & 0 & 0 & 28 & 19 & 2 & 3 & 24 & 9 & 14 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 117 factors with signed exponents of absolute value up to 237209; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 23$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (2 \ 0 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 5: character b , column e_2

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (s - 877)y^3 + (-12s - 96416)y^2 + (-415s + 375030)y + (-1650s + 11227340)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned}
& y^{10} - 4y^9 - 1749y^8 - 189338y^7 + 28568659y^6 - 449980264y^5 \\
& - 9697008411y^4 + 85627040190y^3 + 3623855094940y^2 + 44933719337 \\
& 800y + 198630302329600
\end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 0 \ -1 \ 0 \ -1 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{7063831}{103621065612199888} + \left(\frac{88117}{207242131224399776}\right)s, \quad J = \begin{pmatrix} 2906 & 1111 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2906$. **Checked:** $(a')J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $52061859799040 = 2^{13} \cdot 5 \cdot 7 \cdot 13 \cdot 17 \cdot 19 \cdot 83 \cdot 521$:

$$\begin{pmatrix}
50841659960 & 12130262112 & 37652619184 & 26851228606 & 47114686780 & 36745508632 & 17341787248 & 39892124230 & 40486669015 & 7624246489 \\
0 & 8 & 4 & 4 & 1 & 0 & 4 & 6 & 1 & 7 \\
0 & 0 & 4 & 2 & 2 & 0 & 0 & 1 & 1 & 2 \\
0 & 0 & 0 & 2 & 1 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 4 & 0 & 0 & 2 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 4 & 0 & 1 & 3 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 120 factors with signed exponents of absolute value up to 358157; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 23$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \ 0 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 6: character b , column e_3

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (s - 877)y^3 + (-12s - 96416)y^2 + (-415s + 375030)y + (-1650s + 11227340)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned}
& y^{10} - 4y^9 - 1749y^8 - 189338y^7 + 28568659y^6 - 449980264y^5 \\
& - 9697008411y^4 + 85627040190y^3 + 3623855094940y^2 + 44933719337 \\
& 800y + 198630302329600
\end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 1 \ 0 \ -1 \ 0 \ -1 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{209517721}{100421421980737824} + \left(-\frac{46037}{100421421980737824}\right)s, \quad J = \begin{pmatrix} 2514 & 1788 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2514$. **Checked:** $(a') J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $540412579840 = 2^{12} \cdot 5 \cdot 7^3 \cdot 19 \cdot 4049$:

$$\begin{pmatrix} 603139040 & 265790784 & 213675672 & 595758800 & 106712749 & 211161735 & 542220938 & 219612094 & 369168465 & 269438005 \\ 0 & 16 & 0 & 13 & 1 & 10 & 6 & 6 & 14 & 7 \\ 0 & 0 & 56 & 20 & 24 & 50 & 55 & 35 & 13 & 23 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 126 factors with signed exponents of absolute value up to 569085; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 23$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \ 0 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 7: character c , column e_1

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s - 1510)y^3 + (-23s + 41674)y^2 + (-176s + 1000416)y + (-700s + 7335720)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 3020y^8 + 86346y^7 + 30864015y^6 + 1113401064y^5 \\ & + 22191032252y^4 + 314403718160y^3 + 3296468512064y^2 + 212458261 \\ & 74720y + 66873539174400 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 1 \ 0 \ 0 \ -1 \ 0 \ -1 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{628074433}{495168272762461299} + \left(\frac{61462}{495168272762461299}\right)s, \quad J = \begin{pmatrix} 3459 & 2637 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3459$. **Checked:** $(a') J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $93791761116912 = 2^4 \cdot 3^8 \cdot 271 \cdot 3296897$:

$$\begin{pmatrix} 96493581396 & 18390163599 & 44798973900 & 31195566558 & 36712657159 & 58877142214 & 26863201002 & 88581604925 & 1910607782 & 86381622132 \\ 0 & 3 & 0 & 0 & 2 & 0 & 2 & 2 & 1 & 1 \\ 0 & 0 & 3 & 0 & 0 & 0 & 2 & 0 & 0 & 1 \\ 0 & 0 & 0 & 54 & 53 & 39 & 50 & 15 & 21 & 23 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 93 factors with signed exponents of absolute value up to 53323760403; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \ 4 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 8: character c , column e_2

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s - 1510)y^3 + (-23s + 41674)y^2 + (-176s + 1000416)y + (-700s + 7335720)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 3020y^8 + 86346y^7 + 30864015y^6 + 1113401064y^5 \\ & + 22191032252y^4 + 314403718160y^3 + 3296468512064y^2 + 212458261 \\ & 74720y + 66873539174400 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 1 \ 0 \ 0 \ -1 \ 0 \ -1 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{7063831}{103621065612199888} + \left(\frac{88117}{207242131224399776}\right)s, \quad J = \begin{pmatrix} 2906 & 1111 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2906$. **Checked:** $(a')J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $240768953510400 = 2^9 \cdot 3^2 \cdot 5^2 \cdot 7 \cdot 1453 \cdot 205487$:

$$\begin{pmatrix} 250800993240 & 137241147000 & 3579948406 & 248832458530 & 86257672909 & 137710451145 & 218540851808 & 130407476222 & 208095132380 & 152103259492 \\ 0 & 60 & 30 & 20 & 23 & 15 & 9 & 50 & 7 & 7 \\ 0 & 0 & 2 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 93 factors with signed exponents of absolute value up to 74800076876; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (1 \ 3 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 9: character c , column e_3

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s - 1510)y^3 + (-23s + 41674)y^2 + (-176s + 1000416)y + (-700s + 7335720)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 3020y^8 + 86346y^7 + 30864015y^6 + 1113401064y^5 \\ & + 22191032252y^4 + 314403718160y^3 + 3296468512064y^2 + 212458261 \\ & 74720y + 66873539174400 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 1 \ 0 \ 0 \ -1 \ 0 \ -1 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{209517721}{100421421980737824} + \left(-\frac{46037}{100421421980737824}\right)s, \quad J = \begin{pmatrix} 2514 & 1788 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2514$. **Checked:** $(a')J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $81961158327552 = 2^8 \cdot 3^4 \cdot 7 \cdot 1019 \cdot 554129$:

$$\begin{pmatrix} 47431225884 & 15771391968 & 35485407492 & 38537250028 & 38260024804 & 41561872238 & 31478083928 & 42114838365 & 43918366874 & 47030527372 \\ 0 & 12 & 0 & 4 & 7 & 4 & 8 & 8 & 5 & 6 \\ 0 & 0 & 12 & 4 & 10 & 11 & 6 & 7 & 8 & 0 \\ 0 & 0 & 0 & 4 & 1 & 3 & 3 & 2 & 1 & 3 \\ 0 & 0 & 0 & 0 & 3 & 0 & 2 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 89 factors with signed exponents of absolute value up to 86121435080; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (4 \ 3 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (s - 1221)y^3 + (-39s - 82872)y^2 + (-1008s + 1676781)y + (40122s + 132676299)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 2440y^8 - 163342y^7 + 31673031y^6 - 1615497435y^5 \\ & - 10685395611y^4 + 3634416891864y^3 - 75544249516236y^2 - 171195 \\ & 8162915520y + 60533089960007295 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ 0 \ 0 \ -1 \ -1 \ -1 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{628074433}{495168272762461299} + \left(\frac{61462}{495168272762461299}\right)s, \quad J = \begin{pmatrix} 3459 & 2637 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3459$. **Checked:** $(a')J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $113744432395125 = 3^3 \cdot 5^3 \cdot 7 \cdot 4814579149$:

$$\begin{pmatrix} 505530810645 & 463941284880 & 267728442435 & 54969227505 & 443218255004 & 108726571234 & 415291919916 & 426573618015 & 291113101741 & 323418708103 \\ 0 & 15 & 0 & 0 & 1 & 11 & 0 & 14 & 8 & 9 \\ 0 & 0 & 5 & 2 & 0 & 4 & 4 & 0 & 4 & 3 \\ 0 & 0 & 0 & 3 & 2 & 1 & 1 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 331 factors with signed exponents of absolute value up to 39114723232309; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_1) = (1 \ 4 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 11: character $a + b$, column e_2

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (s - 1221)y^3 + (-39s - 82872)y^2 + (-1008s + 1676781)y + (40122s + 132676299)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 2440y^8 - 163342y^7 + 31673031y^6 - 1615497435y^5 \\ & - 10685395611y^4 + 3634416891864y^3 - 75544249516236y^2 - 171195 \\ & 8162915520y + 60533089960007295 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ 0 \ 0 \ -1 \ -1 \ -1 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{7063831}{103621065612199888} + \left(\frac{88117}{207242131224399776}\right)s, \quad J = \begin{pmatrix} 2906 & 1111 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2906$. **Checked:** $(a')J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $683444964983985 = 3^2 \cdot 5 \cdot 15187665888533$:

$$\begin{pmatrix} 227814988327995 & 13495041524460 & 130174409700515 & 212468347312785 & 219157139311409 & 20034470865309 & 209317044922506 & 89746304750640 & 113370583763036 & 167381246951893 \\ 0 & 3 & 1 & 0 & 1 & 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 334 factors with signed exponents of absolute value up to 144565173857909; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_2) = (3 \ 2 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 12: character $a + b$, column e_3

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (s - 1221)y^3 + (-39s - 82872)y^2 + (-1008s + 1676781)y + (40122s + 132676299)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 2440y^8 - 163342y^7 + 31673031y^6 - 1615497435y^5 \\ & - 10685395611y^4 + 3634416891864y^3 - 75544249516236y^2 - 171195 \\ & 8162915520y + 60533089960007295 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ 0 \ 0 \ -1 \ -1 \ -1 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{209517721}{100421421980737824} + \left(-\frac{46037}{100421421980737824}\right)s, \quad J = \begin{pmatrix} 2514 & 1788 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2514$. **Checked:** $(a')J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $87534615181419 = 3^5 \cdot 7 \cdot 23971 \cdot 2146789$:

$$\begin{pmatrix} 3242022784497 & 3120551347569 & 1781573678223 & 768605153481 & 452115395216 & 220472116784 & 1109362670874 & 2253096761205 & 1539409041683 & 2688253619641 \\ 0 & 3 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 2 & 2 & 2 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 339 factors with signed exponents of absolute value up to 218099500408625; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \ 0 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3y^4 + (s + 1797)y^3 + (-16s - 219447)y^2 + (-263s + 2983268)y + (2546s - 7929745)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 6y^9 + 3604y^8 - 449695y^7 + 37179083y^6 - 1675978848y^5 \\ & + 51732873662y^4 - 977639481840y^3 + 12050928620806y^2 - 8301318 \\ & 8530561y + 235706769450839 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 0 \ -1 \ 0 \ -1 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{628074433}{495168272762461299} + \left(\frac{61462}{495168272762461299}\right)s, \quad J = \begin{pmatrix} 3459 & 2637 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3459$. **Checked:** $(a')J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1346413176596005 = 5 \cdot 7 \cdot 7159 \cdot 5373508577$:

$$\begin{pmatrix} 1346413176596005 & 550005499012546 & 913638317690649 & 690599832420529 & 1101271366357053 & 148222726627857 & 876544696714855 & 302809864642500 & 502024132418327 & 63876572328771 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 386 factors with signed exponents of absolute value up to 6995885778952; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 3 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3y^4 + (s + 1797)y^3 + (-16s - 219447)y^2 + (-263s + 2983268)y + (2546s - 7929745)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 6y^9 + 3604y^8 - 449695y^7 + 37179083y^6 - 1675978848y^5 \\ + 51732873662y^4 - 977639481840y^3 + 12050928620806y^2 - 8301318 \\ 8530561y + 235706769450839$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 0 \ -1 \ 0 \ -1 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{7063831}{103621065612199888} + \left(\frac{88117}{207242131224399776}\right)s, \quad J = \begin{pmatrix} 2906 & 1111 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2906$. **Checked:** $(a')J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $113431576876625 = 5^3 \cdot 7 \cdot 17 \cdot 6007 \cdot 1269461$:

$$\begin{pmatrix} 4537263075065 & 3508820104315 & 4457686810690 & 1082990814211 & 4455782888743 & 199314049086 & 777157767647 & 1014504016260 & 2725283420073 & 2764282881801 \\ 0 & 5 & 0 & 1 & 1 & 4 & 1 & 0 & 1 & 3 \\ 0 & 0 & 5 & 4 & 1 & 0 & 4 & 0 & 0 & 3 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 381 factors with signed exponents of absolute value up to 8453124904169; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_2) = (4 \ 0 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3y^4 + (s + 1797)y^3 + (-16s - 219447)y^2 + (-263s + 2983268)y + (2546s - 7929745)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 6y^9 + 3604y^8 - 449695y^7 + 37179083y^6 - 1675978848y^5 \\ + 51732873662y^4 - 977639481840y^3 + 12050928620806y^2 - 8301318 \\ 8530561y + 235706769450839$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ -1 \ 0 \ 0 \ -1 \ 0 \ -1 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{209517721}{100421421980737824} + \left(-\frac{46037}{100421421980737824}\right) s, \quad J = \begin{pmatrix} 2514 & 1788 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2514$. **Checked:** $(a') J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is 151454047292905 = $5 \cdot 7^2 \cdot 17 \cdot 19 \cdot 107 \cdot 17886629$:

$$\begin{pmatrix} 21636292470415 & 2777737321912 & 9444215882468 & 7140550998799 & 20815017408200 & 2985918302308 & 6557805212005 & 18173701127952 & 7781202680913 & 8272009911399 \\ 0 & 7 & 0 & 6 & 1 & 6 & 6 & 2 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 381 factors with signed exponents of absolute value up to 11184926418494; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (4 \ 3 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 4y^4 + (-s + 3222)y^3 + (20s + 282696)y^2 + (1396s - 2087160)y + (-656s - 87692192)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 8y^9 + 6459y^8 + 539640y^7 + 30608564y^6 + 596178464y^5 \\ & + 3398291520y^4 - 220751263744y^3 + 6036551389184y^2 + 3170957521 \\ & 71520y + 7701453005127680 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ 0 \ -1 \ 1 \ 1 \ -1 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{628074433}{495168272762461299} + \left(\frac{61462}{495168272762461299}\right) s, \quad J = \begin{pmatrix} 3459 & 2637 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3459$. **Checked:** $(a') J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $267124184985088 = 2^9 \cdot 17 \cdot 113^2 \cdot 2403463$:

$$\begin{pmatrix} 36936419384 & 17308250680 & 32550810830 & 5215878061 & 3679901483 & 12167536014 & 33412165383 & 25579986456 & 33895112601 & 21296870524 \\ 0 & 452 & 0 & 450 & 37 & 0 & 180 & 134 & 294 & 3 \\ 0 & 0 & 2 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 197 factors with signed exponents of absolute value up to 614977045; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 3 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 4y^4 + (-s + 3222)y^3 + (20s + 282696)y^2 + (1396s - 2087160)y + (-656s - 87692192)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 8y^9 + 6459y^8 + 539640y^7 + 30608564y^6 + 596178464y^5 \\ & + 3398291520y^4 - 220751263744y^3 + 6036551389184y^2 + 3170957521 \\ & 71520y + 7701453005127680 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ 0 \ -1 \ 1 \ 1 \ -1 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{7063831}{103621065612199888} + \left(\frac{88117}{207242131224399776}\right)s, \quad J = \begin{pmatrix} 2906 & 1111 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2906$. **Checked:** $(a')J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $481904011282900 = 2^2 \cdot 5^2 \cdot 577 \cdot 8351889277$:

$$\begin{pmatrix} 48190401128290 & 15089846386290 & 2593477230179 & 40203499376119 & 25081025194083 & 37609245131089 & 40148273987758 & 30271870955066 & 3734929388947 & 12233685325432 \\ 0 & 10 & 6 & 2 & 3 & 6 & 8 & 5 & 0 & 5 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 195 factors with signed exponents of absolute value up to 1650760; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \ 1 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 4y^4 + (-s + 3222)y^3 + (20s + 282696)y^2 + (1396s - 2087160)y + (-656s - 87692192)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 8y^9 + 6459y^8 + 539640y^7 + 30608564y^6 + 596178464y^5 \\ & + 3398291520y^4 - 220751263744y^3 + 6036551389184y^2 + 3170957521 \\ & 71520y + 7701453005127680 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ 0 \ -1 \ 1 \ 1 \ -1 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{209517721}{100421421980737824} + \left(-\frac{46037}{100421421980737824}\right)s, \quad J = \begin{pmatrix} 2514 & 1788 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2514$. **Checked:** $(a')J^5 = \mathcal{O}_K$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $515077082455040 = 2^{10} \cdot 5 \cdot 7^2 \cdot 17 \cdot 43 \cdot 2808593$:

$$\begin{pmatrix} 4598902521920 & 894729359118 & 3516780975776 & 2415140113855 & 2982561655812 & 4037883890922 & 934544403959 & 3673579685834 & 314779591982 & 4296125929399 \\ 0 & 14 & 6 & 4 & 4 & 4 & 11 & 8 & 5 & 4 \\ 0 & 0 & 2 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 2 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 220 factors with signed exponents of absolute value up to 165798006; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.